Towards Improving the Reliability of AI: Perspectives from Uncertainty Quantification

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Context

- Artificial Intelligence (AI) is emerging as an important technology

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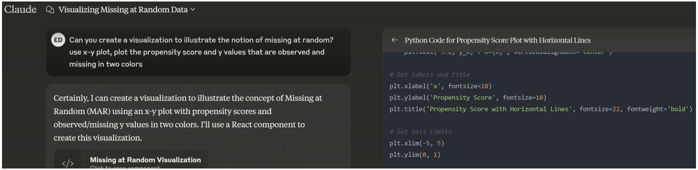
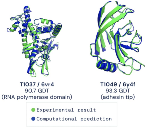
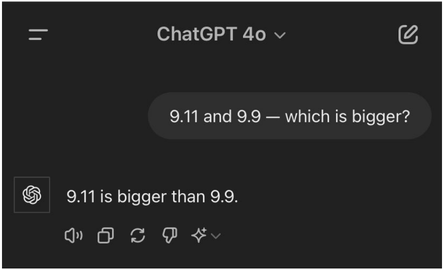
- Success stories: Large language models (ChatGPT, Claude), Image generation (Dall-E, Stable Diffusion), Protein structure prediction ...

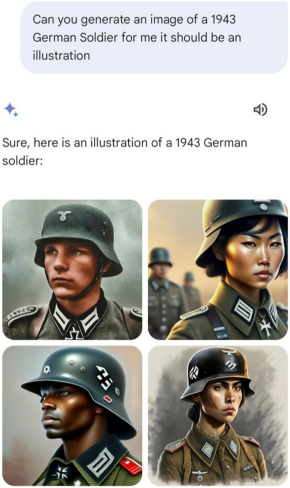




Current AI has problems

- AI systems can be wrong, discriminatory, and racist





Towards reliable AI via uncertainty quantification

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  1. Refrain from taking action when uncertainty is high

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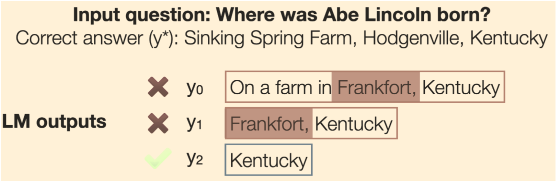

Towards reliable AI via uncertainty quantification

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     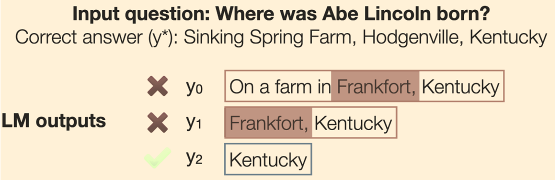

  3. Consider worst-case scenario over outcomes consistent with uncertainty

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Example uncertainty measures

- Many uncertainty measures for LLMs have been discussed/proposed
  - Perplexity: $U(x, y) = \hat{p}(y|x)^{1/\mathrm{len}(y)}$; [related to NLL $-\log \hat{p}(y|x)$]

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  - Semantic entropy: generate multiple ys, cluster them based on meaning, calculate entropy (Kuhn et al., 2023)

Answers to the question "What is the capital of France?"

Answer $\mathbf{s}$	Likelihood $p(\mathbf{s} \mid x)$	Semantic likelihood $\sum_{\mathbf{s} \in c} p(\mathbf{s} \mid x)$
Paris	0.5	0.9
It's Paris	0.4	
London	0.1	0.1
Entropy	0.94	0.33

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  - Affinity graph: generate multiple ys, put pairwise similarities in a matrix, find eigenvalues/vectors (denoted EigV) (Lin et al., 2023)

Uncertainty quantification in AI

- What properties does good uncertainty quantification have?

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Uncertainty in Language Models: Assessment through Rank-Calibration

Xinmeng Huang[*†] **Shuo Li**[*†] **Mengxin Yu**[†] **Matteo Sesia**[‡] **Hamed Hassani**[†]

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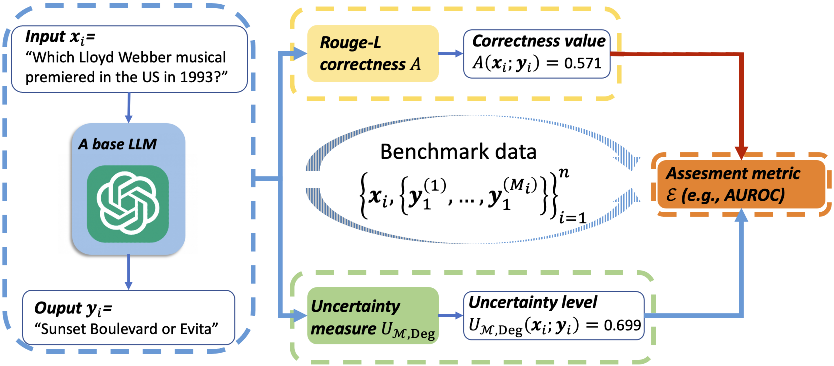
Workflow for using uncertainty measures



Good uncertainty measures

$\boldsymbol{x}$: `On September 28th, NASA announced that what had been detected on Mars?`
$\boldsymbol{y}^*$: `flowing water`
$\boldsymbol{y}$: `Possible signs of life`

Good uncertainty measures

```
x:  On September 28th, NASA announced that
what had been detected on Mars?
y*:  flowing water
y:  Possible signs of life
```

- Semantic Entropy: 0.813
- Perplexity: 1.432
- How to compare (different scales)?

Good uncertainty measures: transform to percentiles

$\boldsymbol{x}$: `On September 28th, NASA announced that what had been detected on Mars?`
$\boldsymbol{y}^*$: `flowing water`
$\boldsymbol{y}$: `Possible signs of life`

- Semantic Entropy: 51.3rd percentile
- Perplexity: 93rd percentile
- Which one is more informative?

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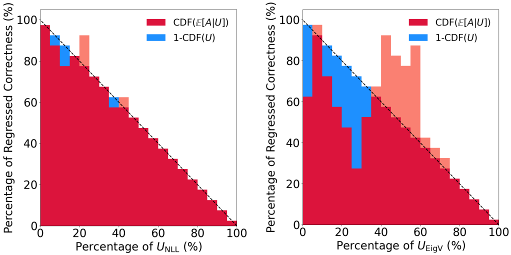

Figure: U_{NLL} (negative log-likelihood) and U_{EigV}, for the GPT-3.5-turbo model on the TriviaQA benchmark.

Good uncertainty measures: rank-calibration

- Our idea: *performance should decrease as a function of the uncertainty*

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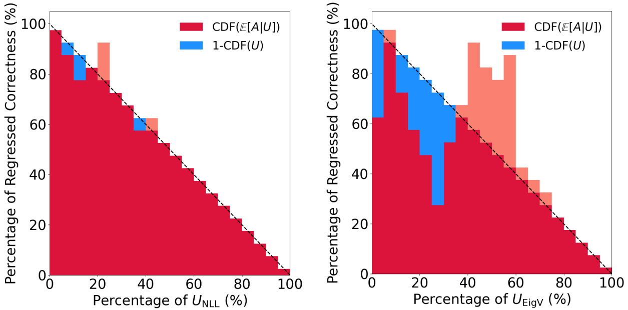

Figure: *Indication diagrams* for U_{NLL} (negative log-likelihood) and U_{EigV}, for the GPT-3.5-turbo model on the TriviaQA benchmark.

Regression function and indication diagram

- Define the regression function $\mathrm{reg}(\cdot) : \mathbb{R} \to \mathbb{R}$, $u \mapsto \mathbb{E}_{\boldsymbol{x},\boldsymbol{y}^*}[A(\boldsymbol{x};\boldsymbol{y}^*) \mid U(\boldsymbol{x};\boldsymbol{y}^*) = u]$, representing the *expected correctness level A given an uncertainty level $U = u$*.

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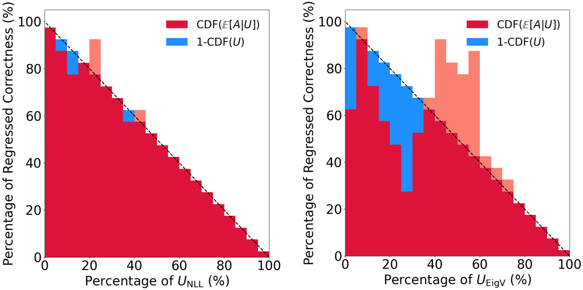
- Indication diagram: plot of estimated regression function as a fn. of U-percentiles.



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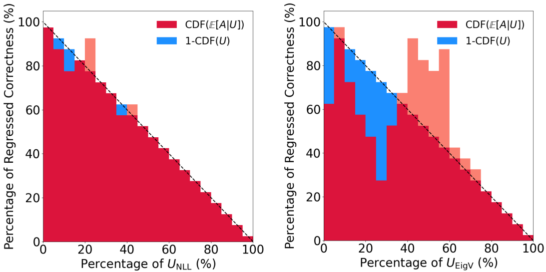
Rank-calibration

Definition (Rank-Calibration)

An uncertainty measure U is *rank-calibrated* if reg is strictly *monotone decreasing*: on average, lower uncertainty implies higher generative quality.



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Rank-calibration error (RCE)

Rank-calibration implies $\mathbb{P}(U \leq u') = \mathbb{P}(\mathrm{reg}(U) \geq \mathrm{reg}(u'))$ for all $u' \in \mathrm{dom}(U)$.

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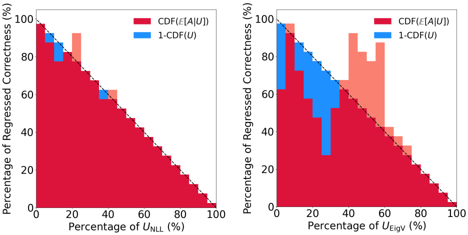
Definition (Rank-Calibration Error)

The RCE of an uncertainty measure U is defined as

$$\mathbb{E}_{U'}\left[\left|\mathbb{P}_U(U \leq U') - \mathbb{P}_U(\mathrm{reg}(U) \geq \mathrm{reg}(U'))\right|\right],$$

where U' is an independent copy of U.



Empirical results

correctness	temperature	U_{Deg}	U_{EigV}	U_{NLL}	U_{SE}
bert	0.5	$0.212_{\pm 0.040}$	$0.212_{\pm 0.041}$	$\mathbf{0.043}_{\pm 0.006}$	$0.052_{\pm 0.009}$
	1.0	$0.129_{\pm 0.020}$	$0.133_{\pm 0.020}$	$\mathbf{0.039}_{\pm 0.007}$	$0.052_{\pm 0.012}$
	1.5	$0.053_{\pm 0.011}$	$0.074_{\pm 0.012}$	$\mathbf{0.031}_{\pm 0.007}$	$0.081_{\pm 0.009}$
meteor	0.5	$0.211_{\pm 0.045}$	$0.208_{\pm 0.047}$	$\mathbf{0.179}_{\pm 0.021}$	$0.234_{\pm 0.019}$
	1.0	$0.131_{\pm 0.024}$	$\mathbf{0.131}_{\pm 0.022}$	$0.146_{\pm 0.011}$	$0.209_{\pm 0.012}$
	1.5	$\mathbf{0.059}_{\pm 0.011}$	$0.077_{\pm 0.012}$	$0.119_{\pm 0.010}$	$0.176_{\pm 0.015}$
rougeL	0.5	$0.210_{\pm 0.042}$	$0.207_{\pm 0.041}$	$\mathbf{0.041}_{\pm 0.007}$	$0.050_{\pm 0.008}$
	1.0	$0.126_{\pm 0.019}$	$0.129_{\pm 0.019}$	$\mathbf{0.038}_{\pm 0.007}$	$0.059_{\pm 0.009}$
	1.5	$0.059_{\pm 0.012}$	$0.079_{\pm 0.011}$	$\mathbf{0.034}_{\pm 0.008}$	$0.104_{\pm 0.007}$
rouge1	0.5	$0.212_{\pm 0.043}$	$0.209_{\pm 0.042}$	$\mathbf{0.040}_{\pm 0.007}$	$0.050_{\pm 0.008}$
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	1.5	$0.060_{\pm 0.011}$	$0.078_{\pm 0.012}$	$\mathbf{0.034}_{\pm 0.008}$	$0.105_{\pm 0.008}$

Table: RCE results for various experimental configurations, for the GPT-3.5-turbo model on the TriviaQA benchmark.

Finding: models often have large rank-calibration error (so uncertainty measures do not reflect performance).

Reducing RCE via re-calibration

- Re-calibrate uncertainty measure: change it to piece-wise constant estimate of regression function; Inspired by classical re-calibration

(Mincer-Zarnowitz, 1969)

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- Improves RCE:

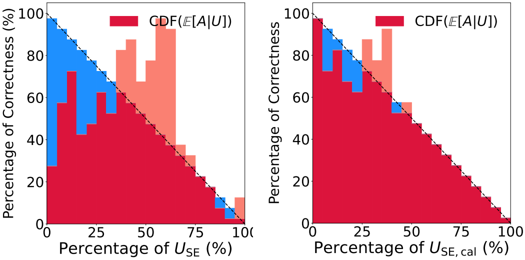

Figure: Indication diagrams of U_{SE} and $U_{\mathrm{SE,cal}}$ (re-calibrated) for GPT-3.5-turbo (temperature 1.0) on TriviaQA with the Meteor correctness.

Summary for rank-calibration

- Proposed rank-calibration for LLMs: higher uncertainty should imply lower quality (not satisfied by default!)

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- Future directions:
  1. How does rank-calibration improves performance?
  2. Better metrics of miscalibration

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Conformal prediction

- Major developing area: conformal prediction/distribution-free predictive inference

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- Goal, given $(X_1, Y_1), \ldots, (X_n, Y_n)$, find a prediction set C such that for new X_{n+1}, $\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha$ under *minimal assumptions*

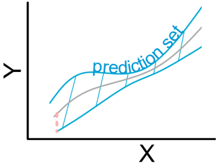

Figure: Towards DS

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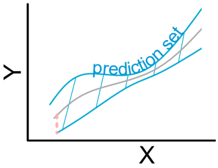

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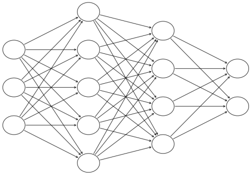

- Motivation: a machine learning model $\hat{\mu}$ is used to predict Y_{n+1} based on X_{n+1}. Not known how to find distribution of $Y_{n+1} - \hat{\mu}(X_{n+1})$

Conformal prediction

- It is known how to do this in many settings, due to extensive work by many, starting in the 90s (Vovk, Wasserman, J. Lei, R. J. Tibshirani, Barber, Candes, ...)
- Ideas date back to work on tolerance regions by Wilks, Wald, Tukey ... starting in the 1940s

Samuel S. Wilks

Abraham Wald

Vladimir Vovk

Conformal prediction ctd.

- Typical setting: *exchangeable datapoints.*
  - For a given nonconformity score $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, e.g., $s(x, y) := |y - \hat{\mu}(x)|$, $s(X_1, Y_1), \ldots, s(X_{n+1}, Y_{n+1})$ are exchangeable (if $\hat{\mu}$ is pre-trained on an indep. dataset—i.e., in split conformal prediction)

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  - Hence, the rank of $s(X_{n+1}, Y_{n+1})$ is uniform over $1, \ldots, n+1$ (if no ties)
  - So $x \mapsto C(x) = \{y : \mathrm{rank}\{s(x, y) : s_1, \ldots, s_n\} \leq \lceil (1-\alpha)(n+1) \rceil\}$ satisfies $\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha$

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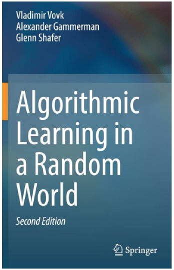
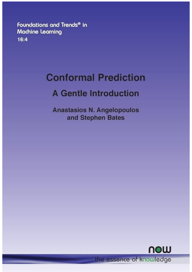




Simultaneous Conformal Prediction of Missing Outcomes with Propensity Score ε-Discretization

Yonghoon Lee

Eric Tchetgen Tchetgen

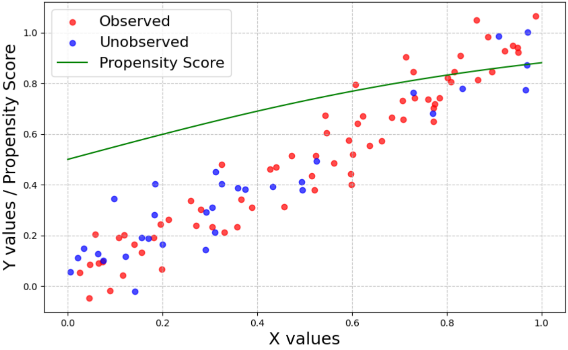
Our problem setting

- Given data

$$(X_1, A_1, Y_1 A_1), \ldots, (X_n, A_n, Y_n A_n) \overset{\text{iid}}{\sim} P_X \times P_{A|X} \times P_{Y|X},$$

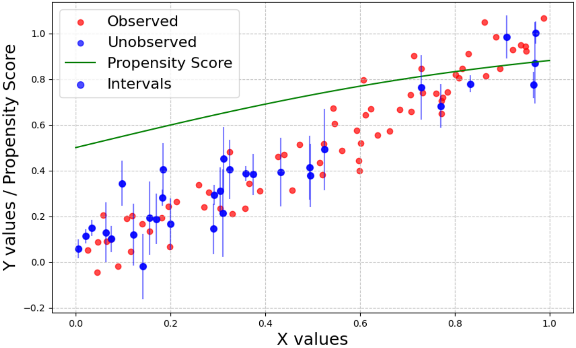
with outcomes *missing at random* (MAR). Thus,

$$A_i = 1 : Y_i \text{ is observed}, \qquad A_i = 0 : Y_i \text{ is unobserved}.$$



Our problem setting: Missing At Random

- **Goal**: Construct prediction sets $\{\widehat{C}(X_i) : A_i = 0\}$ for the missing outcomes $\{Y_i : A_i = 0\}$



Inferential target

- With i.i.d./exchangeable data $(X_1, Y_1), \cdots, (X_n, Y_n)$ and test input X_{n+1}, standard conformal prediction gives a prediction set $\widehat{C}_n(X_{n+1})$ with *marginal coverage*

$$\mathbb{P}(Y_{n+1} \in \widehat{C}_n(X_{n+1})) \geq 1 - \alpha.$$

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- **Question:** Under MAR:
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- **Question:** Under MAR:
  - What kind of distribution-free inference is possible for missing outcomes?
  - Is it possible to go beyond marginal coverage? E.g., have coverage conditional on the test inputs/feature observations with missing outcomes?

Overview of results

- We consider coverage guarantees of the form

$$\mathbb{E}\left[\frac{1}{N^{(0)}}\sum_{i:A_i=0} I\{Y_i \in \widehat{C}(X_i)\}\,\middle|\, X_{1:n}, A_{1:n}\right] \geq 1-\alpha, \tag{1}$$

  where $N^{(0)}$ is the number of unobserved labels, and $0/0 := 1$.
- The proportion of covered missing outcomes is on average at least $1-\alpha$, conditional on $X_{1:n}$ and the missingness pattern $A_{1:n}$.

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- The proportion of covered missing outcomes is on average at least $1-\alpha$, conditional on $X_{1:n}$ and the missingness pattern $A_{1:n}$.
  - For discrete features X, we construct a procedure that achieves (1).
  - For general features X, we prove an impossibility result for (1); and then relax it.

Overview of results - continued

- As a relaxation, we consider

$$\mathbb{E}\left[\frac{1}{N^{(0)}}\sum_{i:A_i=0} I\{Y_i \in \widehat{C}(X_i)\}\,\middle|\, B_{1:n}, A_{1:n}\right] \geq 1-\alpha, \qquad (2)$$

where $B_i = B_i(X_i)$ is a discretization of X_i (defined soon).

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- We introduce a carefully designed propensity score partitioning scheme, and show how it can be used to obtain (2)

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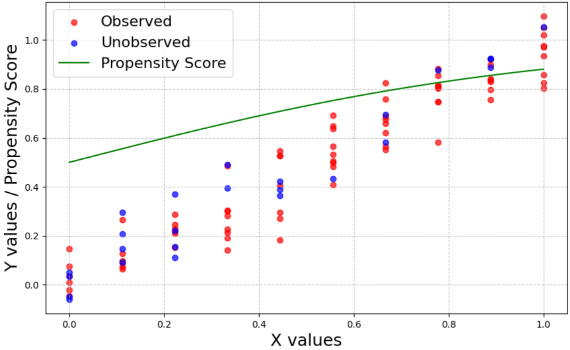
First case: Discrete features

- Discrete features naturally form groups of outcomes $\{Y_i : X_i = x\}$, $x \in \mathcal{X}$.



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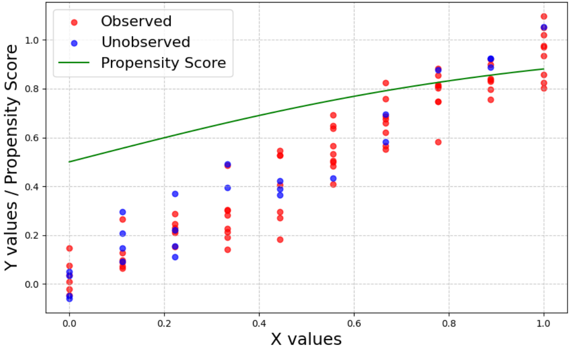
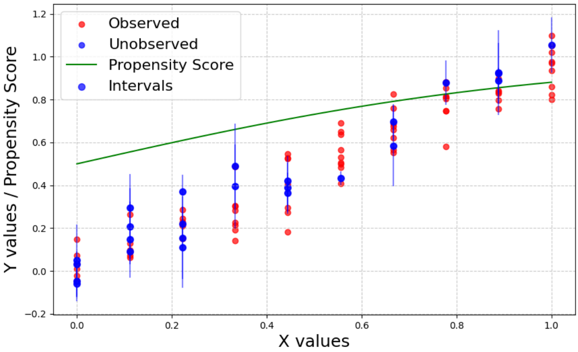


- Within each group, the outcomes are *exchangeable* conditional on $X_i = x$.

Procedure for discrete features: Naive approach

- Direct method: run split conformal prediction separately for each x.



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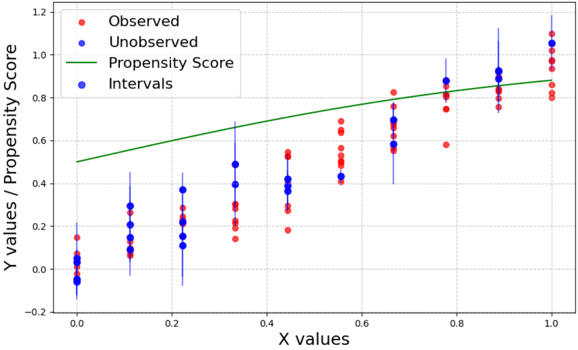

- This method attains $\mathbb{E}\left[\frac{1}{N^{(0)}}\sum_{i:A_i=0} I\{Y_i \in \widehat{C}(X_i)\}\middle| X_{1:n}, A_{1:n}\right] \geq 1-\alpha$.

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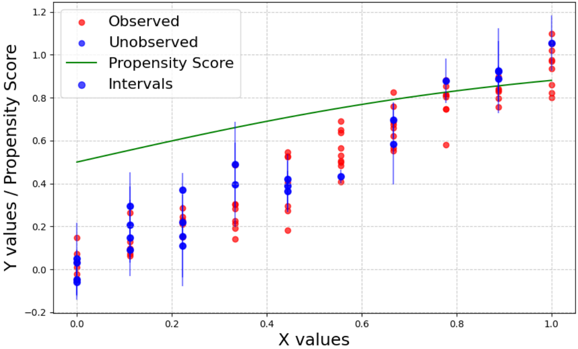

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- However, it can produce infinite-width prediction sets in small groups with $\geq \alpha$ missingness.

Procedure for discrete features: our method

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- Let
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  2. Distinct X values observed: $X'_1, \cdots, X'_M$
  3. Indices of datapoints with features equal to X'_k: $I_k = \{i \in [n] : X_i = X'_k\}$,
  4. Indices partitioned according to unobserved and observed outcomes, resp.: $I_k^0 = \{i \in [n] : X_i = X'_k, A_i = 0\}$, $I_k^1 = \{i \in [n] : X_i = X'_k, A_i = 1\}$.
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- Our prediction set $\widehat{C}(x)$:

$$\left\{ y \in \mathcal{Y} : s(x,y) \le Q_{1-\alpha}\left(\sum_{k=1}^{M} \sum_{i \in I_k^1} \frac{N_k^0}{N_k N^{(0)}} \delta_{S_i} + \sum_{k=1}^{M} \frac{(N_k^0)^2}{N_k N^{(0)}} \delta_{+\infty} \right) \right\}. \qquad (3)$$

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- Idea: conditionally on $X_{1:n}$ and $A_{1:n}$, the distribution of outcomes $Y_{1:n}$ is invariant under the group of permutations that keeps the feature values fixed
- Provides uniform-width prediction sets for all x values.

Procedure for discrete features: guarantee

Theorem 1

The prediction set (3) satisfies *feature- and missingness-conditional coverage*

$$\mathbb{E}\left[\frac{1}{N^{(0)}}\sum_{i:A_i=0} I\{Y_i \in \widehat{C}(X_i)\}\,\middle|\, X_{1:n}, A_{1:n}\right] \geq 1-\alpha.$$

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- Why practically useful? Partition can depend on $X_{1:n}, A_{1:n}$; can aim to ensure small missingness per group.

Procedure for general feature distributions

- If propensity score $x \mapsto p_{A|X}(x) = \mathbb{P}(A = 1|X = x)$ is known, ε-**discretize** it
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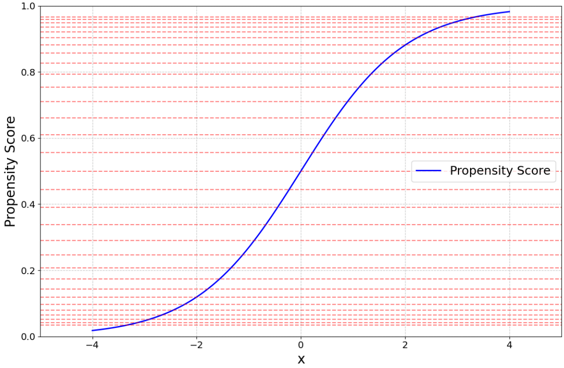

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If $0 < p_{A|X}(X) < 1$ a.s., then $\widehat{C}^{\text{pro-CP}}$ has *propensity score discretized feature- and missingness-conditional coverage*:

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- The error from discretization is $\le \varepsilon$, for *any* n and # of missing outcomes.

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- If the propensity score is unknown, run pro-CP with an estimator $\hat{p}_{A|X}$ of $p_{A|X}$.

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where

$$\delta_{\hat{p}_{A|X}} = e^{2\|\log f_{p,\hat{p}}\|_\infty} - 1, \qquad f_{p,\hat{p}}(x) = \frac{p_{A|X}(x)/(1-p_{A|X}(x))}{\hat{p}_{A|X}(x)/(1-\hat{p}_{A|X}(x))}.$$

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- The error from estimation does not grow with the number of missing outcomes.

New result underlying pro-CP guarantee

- Balancing property of the propensity score [Rosenbaum and Rubin (1983)]: the missingness is independent of the outcome conditional on the propensity: $A \perp\!\!\!\perp Y \mid p_{A|X}$.

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- We show *approximate version*: dist. of $s(X, Y)$ close for $A = 0, 1$ given small range of $p_{A|X}$

Lemma (Bounded prop. score $\Rightarrow$ cond. dist. of obs. and missing are close)

Suppose that $(X, Y, A) \sim P_X \times P_{Y|X} \times \text{Bernoulli}(p_{A|X})$ *on* $\mathcal{X} \times \mathcal{Y} \times \{0, 1\}$, *and that for a set* $B \subset \mathcal{X}$ *and* $t \in (0, 1)$, $\varepsilon \geq 0$,

$$t \leq \frac{p_{A|X}(x)}{1 - p_{A|X}(x)} \leq t(1 + \varepsilon), \text{ for all } x \in B.$$

Let $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ *be any measurable function and let* $S = s(X, Y)$. *Then*

$$d_{TV}(P_{S|A=1,X\in B}, P_{S|A=0,X\in B}) \leq \varepsilon.$$

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Application to simultaneous inference on ITEs

- Consider a potential outcomes model

$$(X_i, T_i, Y_i(0), Y_i(1))_{1\le i\le n} \overset{\text{iid}}{\sim} P_X \times P_{T|X} \times P_{Y(1)|X} \times P_{Y(0)|X},$$

where we observe $(X_i, T_i, T_i Y_i(1) + (1 - T_i) Y_i(0))_{1\le i\le n}$.

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- By letting $\widehat{C}_i^{\text{ITE}} = \{y - Y_i(0) : y \in \widehat{C}^{\text{counterfactual}}(X_i)\}$, we obtain prediction sets for individual treatment effects

$$\mathbb{E}\left[\tfrac{1}{N^{(0)}} \textstyle\sum_{i\in I_{T=0}} \mathbf{1}\left\{(Y_i(1) - Y_i(0)) \in \widehat{C}_i^{\text{ITE}}\right\} \,\middle|\, B_{1:n}, T_{1:n}\right] \ge 1 - \alpha$$

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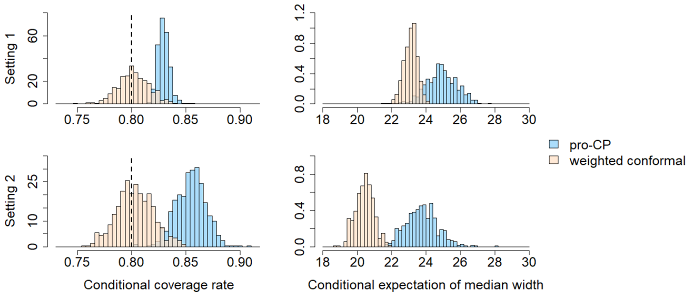


Illustration on diabetes dataset (Efron et al., 2004)

- X: ten features (age, bmi, LDL/HDL, ...) of patients (sample sizes: train: 142; calibration+test: 300)
- A: missingness generated from a known logistic model
- Y: a measure of disease progression one year after baseline

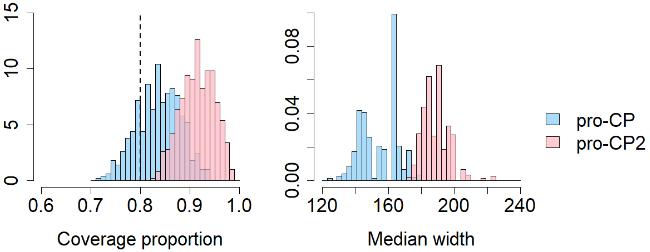


Illustration on diabetes dataset (Efron et al., 2004): II

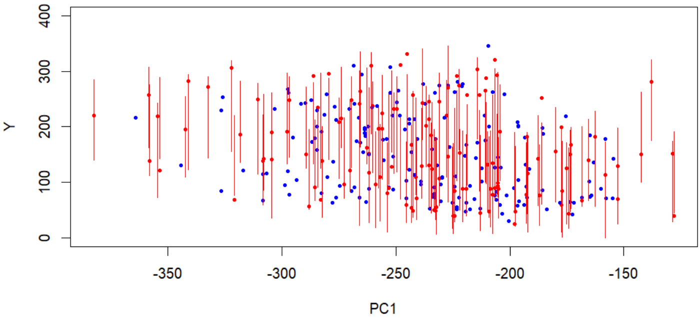
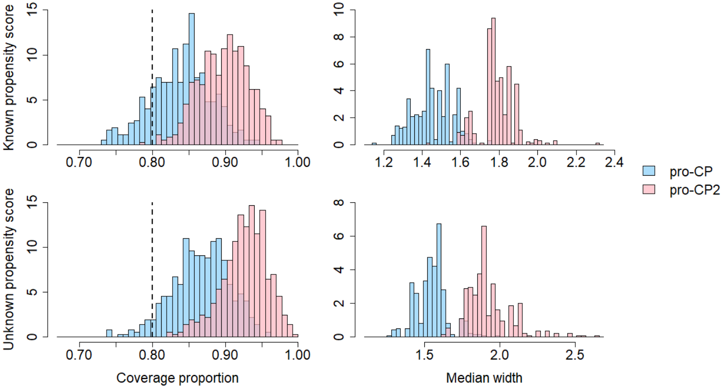


Illustration on JOBS II dataset (Imai et al., 2010)

- X: job seekers: $n_{\text{train}} = 379$, $n = 500$; with 14 demographic features
- A: job skills workshop (to evaluate our methods, simulate via logistic model; estimate via RF)
- $Y(0), Y(1)$: pre- and post-treatment depression measure



Summary for conformal prediction

- Introduced Pro-CP, a method for simultaneous prediction of multiple missing outcomes, and provided coverage guarantees
- Pro-CP2: stronger squared error miscoverage error control

Thank you!

Other Work

Jailbreaking Black Box Large Language Models in Twenty Queries


Patrick Chao, Alexander Robey,
Edgar Dobriban, Hamed Hassani, George J. Pappas, Eric Wong
University of Pennsylvania




`JailbreakBench`: An Open Robustness Benchmark for Jailbreaking Large Language Models

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Leaderboards Paper Contribute Library Behaviors Jailbreak artifacts

JAILBREAKBENCH

Other work (ctd)



One-Shot Safety Alignment for Large Language Models via Optimal Dualization

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Osbert Bastani **Hamed Hassani** **Dongsheng Ding†**

Watermarking Language Models with Error Correcting Codes

Patrick Chao, Edgar Dobriban, Hamed Hassani*

Evaluating the Performance of Large Language Models via Debates

Behrad Moniri **Hamed Hassani** **Edgar Dobriban**